

Factorial Designs

Two-way anova with interactions

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2026-01-01

Learning Objectives

By the end of this lectures (we will probably spend two lectures on this), you should be able to:

- Explain the advantage of factorial designs over one-factor-at-a-time experiments
- Construct the statistical model for a two-factor factorial: $y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$
- Design a factorial experiment using CRD or RCBD structure
- Create and interpret interaction plots to visualize how factors combine
- Conduct simple effects analysis when significant interactions are present
- Make appropriate statistical conclusions when interactions are significant vs non-significant

Motivating Question

Research scenario:

You want to study the effects of **irrigation** and **nitrogen fertilizer** on corn yield.

Two approaches:

1. Run **two separate experiments** (one-factor-at-a-time)
2. Run **one factorial experiment** (both factors together)

Which approach is better?

Review: Single-Factor Designs

CRD (Completely Randomized Design):

- One factor (treatment) with multiple levels
- Randomize treatments to experimental units
- ANOVA: Test treatment effect

Example: Nitrogen (0, 50, 100 kg/ha) on wheat yield

RCBD (Randomized Complete Block Design):

- One factor + blocking for spatial/temporal variation
- Complete set of treatments in each block
- ANOVA: Test treatment effect (accounting for blocks)

Example: Same nitrogen study, blocked by field location

Limitation: Both designs test **one factor at a time**. What if you want to test irrigation AND nitrogen simultaneously?

Why Factorial Designs?

One-Factor-At-a-Time (OFAT) Approach

Experiment 1: Test Irrigation

- Low irrigation (L) vs High irrigation (H)
- Keep nitrogen constant at 50 kg/ha
- 2 treatments $\times$ 8 reps = **16 plots**

Result: High irrigation increases yield by 500 kg/ha

Experiment 2: Test Nitrogen

- No nitrogen (0) vs High nitrogen (100 kg/ha)
- Keep irrigation constant at Low
- 2 treatments $\times$ 8 reps = **16 plots**

Result: Nitrogen increases yield by 300 kg/ha

Problem with OFAT:

- **Total plots:** 32
- **Conclusion:** “Add high irrigation (+500 kg/ha) and nitrogen (+300 kg/ha) for

+800 kg/ha total”

- **Hidden assumption:** Effects are **independent** (additive)
- **Risk:** What if nitrogen only works with high irrigation? **You’ll never know!**

Factorial Solution: Test All Combinations

2×2 Factorial Design:

- Factor A: **Irrigation** (Low, High)
- Factor B: **Nitrogen** (0, 100 kg/ha)
- **Replication:** 8 reps per combination
- **Total plots:** $4 \times 8 = 32$ plots (same as OFAT!)

Treatment combinations: 4 (all possible pairs)

1. Low Irrigation + No Nitrogen (L0)
2. Low Irrigation + High Nitrogen (L100)
3. High Irrigation + No Nitrogen (H0)
4. High Irrigation + High Nitrogen (H100)

Treatment Structure Table:

Irrigation	N = 0	N = 100
Low	L0	L100
High	H0	H100

Factorial advantage:

- Same sample size (32 plots)
- **More information:** Tests irrigation effect, nitrogen effect, AND their interaction!

Factorial Notation

Factorial designs are described by the number of factors and their levels:

- **2×2 factorial:** 2 factors, each with 2 levels → 4 treatment combinations
- **3×3 factorial:** 2 factors, each with 3 levels → 9 treatment combinations
- **2×4 factorial:** 2 factors (2 levels, 4 levels) → 8 treatment combinations
- **2×2×2 factorial:** 3 factors, each with 2 levels → 8 treatment combinations
- **2^k factorial:** k factors, all with 2 levels → 2^k combinations

Examples:

Design	Factors	Levels	Combinations
2×2	Irrigation (2) × Nitrogen (2)	2, 2	4
3×3	Irrigation (3) × Nitrogen (3)	3, 3	9
2×3	Tillage (2) × Variety (3)	2, 3	6
2×2×2	Tillage × Variety × Fungicide	2, 2, 2	8

Main Effects and Interactions

What is a Main Effect?

Main effect of Factor A:

The average effect of Factor A across all levels of Factor B.

- Compares marginal means (averaged over other factors)
- Ignores specific combinations
- Answers: “On average, does Factor A matter?”

Example: Main effect of Irrigation

Mean yield under Low Irrigation:

$$\bar{y}_{\text{Low}} = \frac{y_{L0} + y_{L100}}{2}$$

Average of L0 and L100 (across both nitrogen levels)

Mean yield under High Irrigation:

$$\bar{y}_{\text{High}} = \frac{y_{H0} + y_{H100}}{2}$$

Average of H0 and H100

Main effect of Irrigation:

$$\text{Main Effect}_{\text{Irrigation}} = \bar{y}_{\text{High}} - \bar{y}_{\text{Low}}$$

What is an Interaction?

Interaction between Factor A and Factor B:

The effect of Factor A **depends on** the level of Factor B (or vice versa).

- Main effects are **not additive**
- Simple effects (A at each level of B) differ
- Answers: “Do factors work together in a non-additive way?”

Example: Irrigation × Nitrogen Interaction

If no interaction:

- Nitrogen effect at Low Irrigation = +200 kg/ha
- Nitrogen effect at High Irrigation = +200 kg/ha
- “Nitrogen increases yield by 200 kg/ha regardless of irrigation level”

If interaction present:

- Nitrogen effect at Low Irrigation = +200 kg/ha
- Nitrogen effect at High Irrigation = +1000 kg/ha
- “Nitrogen benefit is much larger under high irrigation!”
- **Factors interact** (synergy)

Biological interpretation:

Nitrogen uptake requires adequate water. Under drought (low irrigation), plants can’t utilize added nitrogen. Under high irrigation, nitrogen is fully effective → **synergistic interaction**.

Interaction Plot

Key observation: Lines are **not parallel** → Interaction present!

- Under **low irrigation** (red): Nitrogen increases yield slightly (3000 → 3200)
- Under **high irrigation** (green): Nitrogen increases yield substantially (3500 → 4500)
- **Interpretation:** Nitrogen benefit depends on irrigation level

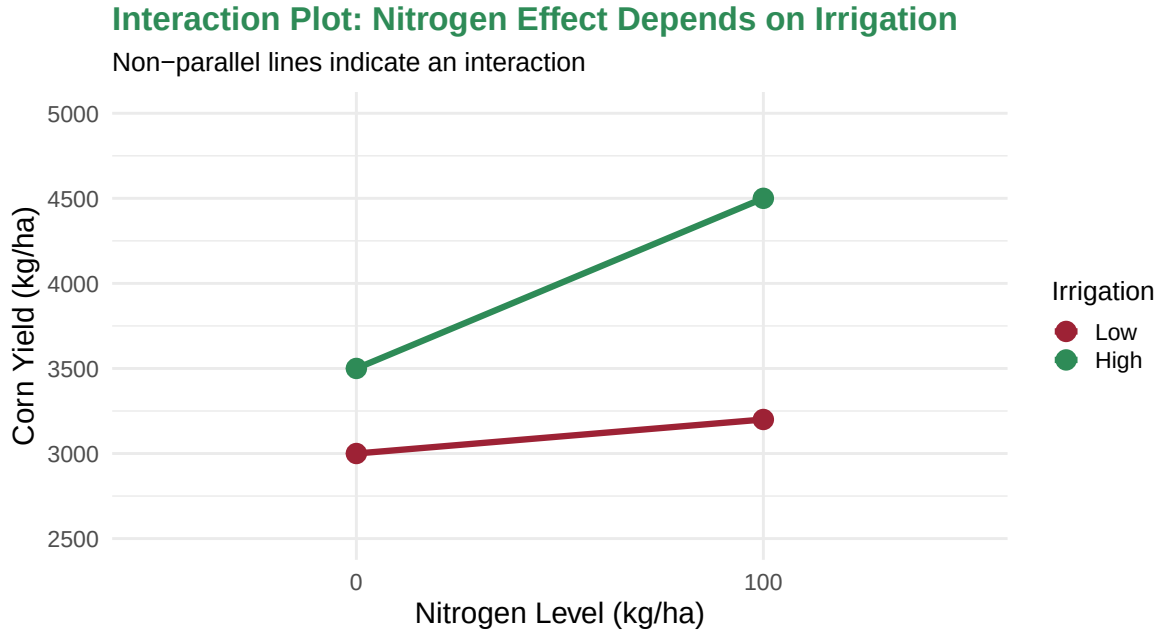


Figure 1: Conceptual interaction plot: Nitrogen effect depends on irrigation

Statistical Model for Factorial Designs

Two-Factor Factorial Model

Mathematical model for 2×2 factorial:

$$y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

Where:

- y_{ijk} = response (e.g., yield) for observation k in treatment i, j
- μ = overall mean (grand mean)
- α_i = main effect of Factor A (e.g., irrigation), $i = 1, 2$
- β_j = main effect of Factor B (e.g., nitrogen), $j = 1, 2$
- $(\alpha\beta)_{ij}$ = interaction effect between Factors A and B
- ε_{ijk} = random error, $\varepsilon_{ijk} \sim N(0, \sigma^2)$

Assumptions

Same assumptions as CRD and RCBD:

1. **Independence:** Observations are independent (proper randomization)
2. **Normality:** Errors ε_{ijk} are normally distributed
3. **Homogeneity of variance:** Equal variance across all treatment combinations (σ^2 constant)
4. **Additivity:** Model is additive (or check for interaction if suspected)

Checking assumptions:

- **Residual plots** (normality, constant variance, outliers)
- **Shapiro-Wilk test** (normality)
- **Levene's test** (homogeneity of variance)

ANOVA for Factorial Design

ANOVA partitions total variation into:

$$SS_{\text{Total}} = SS_A + SS_B + SS_{A \times B} + SS_{\text{Error}}$$

Sources of variation:

1. **Main effect of Factor A** (irrigation)
2. **Main effect of Factor B** (nitrogen)
3. **Interaction A × B** (irrigation × nitrogen)
4. **Error** (within-treatment variation)

ANOVA Table Structure:

Source	df	SS	MS	F	p-value
Factor A	$a - 1$	SS_A	MS_A	$\frac{MS_A}{MS_E}$	
Factor B	$b - 1$	SS_B	MS_B	$\frac{MS_B}{MS_E}$	
A × B	$(a - 1)(b - 1)$	$SS_{A \times B}$	$MS_{A \times B}$	$\frac{MS_{A \times B}}{MS_E}$	
Error	$ab(r - 1)$	SS_E	MS_E		
Total	$abr - 1$	SS_{Total}			

Hypotheses Tested

Main effect of Factor A (Irrigation):

$$H_0 : \alpha_1 = \alpha_2 = 0$$

$$H_A : \text{At least one } \alpha_i \neq 0$$

“Is there an average difference between irrigation levels?”

Main effect of Factor B (Nitrogen):

$$H_0 : \beta_1 = \beta_2 = 0$$

$$H_A : \text{At least one } \beta_j \neq 0$$

“Is there an average difference between nitrogen levels?”

Interaction A×B (Irrigation × Nitrogen):

$$H_0 : (\alpha\beta)_{ij} = 0 \text{ for all } i, j$$

$$H_A : \text{At least one } (\alpha\beta)_{ij} \neq 0$$

“Does the effect of nitrogen depend on irrigation level (or vice versa)?”

Designing a Factorial Experiment

Factorial + CRD Structure

Factorial CRD:

- Treatment structure: Factorial (e.g., $2 \times 2 = 4$ combinations)
- Design structure: Completely randomized (no blocks)
- Randomize all treatment combinations to experimental units

When to use:

- Homogeneous experimental environment
- No obvious blocking factor

Example R code with FieldHub:

```
library(FieldHub)

# Generate 2x2 factorial CRD
design_factorial_crd <- FieldHub::CRD(
  t = 4,                      # 4 treatment combinations (2x2)
  reps = 8,                   # 8 replicates
  plotNumber = 101,
  seed = 2026,
  locationName = "Research Farm"
)

# View layout
print(design_factorial_crd)
plot(design_factorial_crd)
```

Factorial + RCBD Structure

Factorial RCBD:

- Treatment structure: Factorial (e.g., $2 \times 2 = 4$ combinations)
- Design structure: Randomized complete block
- Each block contains all 4 treatment combinations (complete)

When to use:

- Heterogeneous environment (field gradient, time variability)
- Want to control for blocking factor

Example R code with FieldHub:

```
library(FieldHub)

# Generate 2x2 factorial RCBD
design_factorial_rcbd <- FieldHub::RCBD(
  t = 4,                      # 4 treatment combinations (2x2)
  reps = 8,                   # 8 blocks
  plotNumber = 101,
  seed = 2026,
  locationName = "Field Trial"
)
```

```
# View layout
print(design_factorial_rcbd)
plot(design_factorial_rcbd)
```

Design Considerations

Key design decisions for factorials:

1. How many factors?

- Start with 2-3 factors (more factors = exponential growth in plots)
- Example: $2 \times 2 \times 2 = 8$ combinations; $3 \times 3 \times 3 = 27$ combinations

2. How many levels per factor?

- 2 levels: Compare two conditions (yes/no, low/high)
- 3+ levels: Dose-response, multiple varieties

3. Replication:

- Need sufficient power: 6-10 reps per combination typical
- Power analysis: Estimate from pilot data or literature

4. Design structure:

- CRD if environment homogeneous
- RCBD if spatial/temporal variation expected

Your Turn: Design a Factorial

Activity (3 minutes)

Scenario:

You want to study the effects of **tillage method** (Conventional, No-till) and **cover crop** (None, Rye) on soybean yield.

Questions:

1. How many factors? How many levels per factor?
2. What are the treatment combinations?
3. Is this a 2×2 factorial?
4. If you have 6 blocks (fields), how many total plots?

5. Would you use CRD or RCBD?

Discuss with a partner, then we'll share answers.

Activity: Solutions

Answers:

1. **2 factors:** Tillage (2 levels), Cover crop (2 levels)
2. **Treatment combinations:**
 - Conventional + None
 - Conventional + Rye
 - No-till + None
 - No-till + Rye
3. **Yes, 2×2 factorial**
4. **Total plots:** 6 blocks × 4 combinations = **24 plots**
5. **RCBD** (blocks = fields)

Factorial structure:

Tillage	None	Rye
Conventional	Conv-None	Conv-Rye
No-till	NoTill-None	NoTill-Rye

Potential interaction:

Cover crops might improve soil health more under no-till (less soil disturbance) → **tillage × cover crop interaction**

Resources

Required reading: - Oehlert (2010), *A First Course in Design and Analysis of Experiments*, Chapter 8

R packages:

- FieldHub (experimental design generation)
- car (Type III ANOVA)

- `emmeans` (estimated marginal means)
- `ggplot2` (visualization)

Key functions:

- `emmeans(model, ~ FactorB | FactorA)` → Simple effects
- `pairs(emm_object)` → Pairwise comparisons
- `emmip(model, FactorA ~ FactorB, CIs = TRUE)` → Interaction plot with CIs

Additional resources:

- Montgomery (2017), *Design and Analysis of Experiments*, Chapter 5
- Agronomic examples: `agridat` package datasets